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Review Article

Advances in the application of gait analysis in objective evaluation of myelopathy in degenerative cervical spondylosis: A literature review

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ABSTRACT

Objective: Degenerative cervical diseases are the most common cause of non-traumatic spinal cord dysfunction in adults, with degenerative cervical myelopathy (DCM) being the most severe form. Gait analysis is an objective and effective tool for assessing gait function in patients with cervical myelopathy, which can not only support diagnosis and assess disease severity but also evaluate surgical outcomes and post-surgical recovery. This paper reviews the progress of gait analysis in the objective evaluation of cervical myelopathy.

Methods: A search was conducted in databases including CNKI, VIP, Wanfang, Embase, and PubMed, covering literature from January 2015 to September 2024. The Search terms included ((gait) OR (gait analysis) AND ((Degenerative Cervical Myelopathy) OR (Ossification of the Posterior Longitudinal Ligament DCM) OR (degenerative disc disease) OR (ossification of the ligamentum flavum) OR (cervical)). Case reports were excluded.

Results: A total of 98 articles were identified. The studies included clinical research comparing gait patterns in DCM patients and healthy controls, series of studies evaluating gait changes before and after surgical decompression, and studies exploring prognostic prediction through gait analysis, with 10 articles being included in the final review.

1. Introduction

Gait analysis quantifies kinematic and dynamic parameters during walking, revealing the biomechanical and neural control mechanisms of movement. With advancements in motion capture and wearable sensors, its accuracy and clinical applicability have greatly improved [1] In cervical myelopathy, gait analysis can identify abnormalities such as positive straight-leg raise tests and gait instability, providing an objective basis for early diagnosis and rehabilitation, addressing the limitations of traditional imaging in detecting dynamic compression.

Degenerative cervical myelopathy (DCM) is the leading cause of spinal cord injury in individuals over 55, with rising prevalence in younger populations [2]. Early intervention is crucial, but current evaluation methods, including MRI and subjective scales like the JOA score, have significant limitations in sensitivity and accuracy [3,4]. Some studies have explored electromyography (EMG) as an auxiliary tool, but its clinical application remains limited. Therefore, objective technologies like gait analysis are essential for improving surgical decisions and balancing intervention timing [5].

Gait analysis uses wearable and environmental sensors to collect data [6], improving the accuracy of neurodegenerative disease analysis and providing a foundation for quantifying spinal cord function [7,8]. Research on gait in DCM focuses on three areas: gait differences between patients and healthy individuals [9], evaluating functional changes pre- and post-surgery, and exploring prognostic indicators. These studies are transforming spinal cord function assessment from subjective to objective, with potential for reshaping DCM diagnosis and treatment [10,11].

2. Methods

A search was conducted in CNKI, VIP, Wanfang, Embase, and PubMed databases, covering literature from January 2015 to September 2024. The search terms included ((gait) OR (gait analysis) AND ((Degenerative Cervical Myelopathy) OR (Ossification of the Posterior Longitudinal Ligament DCM) OR (degenerative disc disease) OR (ossification of the ligamentum flavum) OR (cervical)). Case reports were excluded.

In this study, we included clinical research that compared the gait of

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DCM patients with healthy controls, studies evaluating gait changes before and after spinal decompression surgery, and studies exploring prognostic indicators using gait analysis. All retrieved articles were carefully reviewed, and references were meticulously screened to identify other relevant studies. We also endeavored to identify any potential duplicate cases across the reported series.

Evidence table reporting study characteristics for all included studies

Author(year)	Study design	LOE	Subgrouping method	Sample size		Age, mean ± SD(years)		Follow-up general (months)	Surgical treatment	Risk of bias (Number of stars in NOS)	System for gait analysis	Outcomes assessed
				cases	controls	case	control					
Zawar Khan Khattak et al 2023	Prospective Cohort	2		10	10	63.6 ± 10.7	69.4 ± 7.0		None	6	wearable tri-accelerometers (Delsys, Inc, Natick, MA, USA)	Average Speed、Cadence、Step length、Stride time、root mean square、root mean square ratio、harmonic ratio、step regularity、stride regularity
Ram Haddas et al 2018	Prospective Cohort	2		28	30	62.20 ± 9.9	57.01 ± 6.1		None	6	Vicon Nexux 2.3, Englewood, CO、force platforms (AMTI)	Lower Extremity Angles at Initial Contact and Throughout the Stance Phase(Ankle、Knee、Hip、Pelvis Orientation、Lumbar Spine、Cervical Spine、Head Orientation)、Walking Speed、Cadence、Step Length、Stride Length、Step Width、Stride Time、Single Support Time、Double Support Time
Ram Haddas et al 2018	Prospective Cohort	2	a:patients with Adult Degenerative Scoliosis (ADS), b:patients with Cervical Spondylotic Myelopathy (CSM)	ADS:20 CSM:20	15	ADS:62.71 ± 6.9、CSM:60.90 ± 6.8	55.01 ± 6.1		None	5	Vicon Video system (Vicon Nexus 2.0 Inc., Englewood, CO)、GRF (AMTI Corp, Watertown, MA.)、EMG (Delsys, Inc, Natick, MA)	Gait speed、Step length、Stride length、Step width、Cadence、Stride time、Stance time、Swing time、Swing/stance ratio、Double support time、Single support time、sagittal and frontal plane angles of the ankle, knee, hip, pelvic
Zdenek Kadanka Jr et al 2021	Retrospective comparative	3	a:DCM, b: NMDCC (non-myelopathic degenerative cervical cord compression)	DCM: 45、NMDCC: 126	100	DCM:59.3 ± 11.8、NMDCC:58.2 ± 9.9	56.1 ± 13.1		None	6	no	10-m walking/running test: Time/Velocity、Number of steps、Cadence
Ram Haddas et al 2020	Prospective Cohort	2		40	30	61.8 ± 10.2	56.9 ± 12.4		Cervical decompression and fusion	5	three force platforms (AMTI)	Vertical Ground Reaction Force、Anterior–Posterior Ground Reaction Force
Ram Haddas et al 2018	Prospective Cohort	2		25	30	63.5 ± 10.4	57.1 ± 16.0	3 months	Twenty patients underwent anterior procedures, 4 posterior procedures, and 1 combined anterior posterior procedures.	7	Vicon Nexux 2.3, Englewood, CO、、3 force platforms (AMTI)	Walking speed、Cadence、Step time、Single support time、Double support time、Step length、Step width、Lower Extremity and Spine Range of Motion
Filippo Mandelli et al 2023	Systematic Review	1										
Neriman Özkan et al 2021	Retrospective comparative	3	a:Nurick grade 0b:Nurick grade 1c:Nurick grade 2 d: Nurick grade 3	411		62.6 ± 12.1		10 Years	Not mention	5		

(continued on next page)

(*continued*)

Evidence table reporting study characteristics for all included studies											
			e:Nurick grade 4								
Hirosuke Nishimura et al 2015	Cross-sectional study	4		34	98	52.7 ± 16.5	a:52.7 ± 16.5, b:67.0 ± 16.9, c:52.7 ± 15.9, d:58.2 ± 10.8, e:67.0 ± 16.9	None	6	Sheet Type Gait Analyzer Walk Way MW-1000, Anima, Tokyo, Japan	Velocity、Step length、Step width、Step angle、Stance phase、Swing phase、Gait cycle
Ippei Kitade et al 2020	Cross-sectional study	4	Group I (JOA score ≥ 10 points)、Group II (JOA score 7 − 9 points)、Group III (JOA score ≤ 6 points)	42	40	Group I:66.3 ± 8.0、Group II:68.0 ± 9.6、Group III:67.4 ± 8.3	67.4 ± 9.3	None	4	VICON 370; Vicon Peak, Oxford, UK、Motion Analysis Corp., Santa Rosa, CA、LG6-4; Advanced Mechanical Technology, Watertown, MA	Spatiotemporal parameters、Kinematic parameters、Kinetics parameter、The elastic ratios of the muscle–tendon

3. Results

3.1. Gait differences between DCM patients and healthy controls: Four studies were included

Degenerative cervical myelopathy (DCM) is a neurological disorder caused by degenerative changes in the cervical spine or spinal cord compression, typically leading to motor dysfunction and gait abnormalities. Gait is a fundamental form of human movement, reflecting physiological status and functional level. In recent years, more studies have focused on gait differences between DCM patients and healthy individuals, attempting to provide new perspectives for early diagnosis and rehabilitation. These studies use various gait parameters, such as walking speed, step length, and step frequency, to reveal significant changes in motor coordination, balance function, and spinal kinematics in DCM patients.

With the development of wearable sensor technology, methods for assessing gait and balance function have significantly improved. By comparing DCM patients with healthy controls, researchers can more accurately uncover the impact of the disease on gait and motor function. A study by Khattak et al [12]. used wearable sensors to assess gait and balance in 10 DCM patients and compared them with 10 age-matched healthy controls. They found that DCM patients had reduced walking speed, step length, and step frequency, along with longer walking time, indicating impaired dynamic balance. Electromyography (EMG) of the ankle and knee joints also showed significant differences. These findings validated the clinical application value of wearable sensor technology in monitoring gait and balance changes in DCM patients, emphasizing the importance of gait data analysis in treatment evaluation.

Haddas et al [13]. analyzed gait in 28 symptomatic DCM patients and 30 healthy controls, evaluating the kinematic parameters of the spine and lower extremities along with spatiotemporal gait parameters. The study showed that DCM patients exhibited significant anterior pelvic tilt and lumbar lordosis at the initial contact of the gait cycle, while cervical lordosis and head flexion were reduced. Additionally, DCM patients had prolonged double support time, shortened single support time, reduced step length, and increased step width. These gait parameters were significantly correlated with spinal kinematic parameters. Moreover, DCM patients had a smaller knee flexion angle during the standing phase, consistent with findings from Maezawa et al [14]., which indicated that increased knee flexion during the standing phase is associated with postoperative neurological improvement. Haddas et al [13]. emphasized the relationship between abnormal spinal kinematics and lower extremity function and highlighted the importance of quantitative gait analysis in assessing and developing rehabilitation plans for DCM patients, especially those with gait abnormalities.

Furthermore, Haddas et al [15]. compared gait in 20 patients with degenerative scoliosis (ADS), 20 patients with cervical myelopathy (DCM), and healthy controls. The results showed that both ADS and DCM patients exhibited significant changes in walking speed, step length, and step frequency, along with reduced range of motion in the ankle, knee, pelvis, and trunk. EMG analysis showed prolonged muscle activation times and altered ground reaction force (GRF) patterns. These findings suggest that gait analysis provides objective functional gait indicators for evaluating ADS and DCM patients. Therefore, the use of gait analysis in clinical assessments is recommended to better understand the impact of spinal diseases on patients' gait, function, and quality of life.

Haddas et al.'s research further revealed the widespread effects of spinal diseases (such as ADS and DCM) on gait and lower limb function, emphasizing the importance of gait analysis in disease evaluation. A study by Kadanka et al [16]. focused on early-stage cervical myelopathy (NMDCC) and verified gait analysis as a sensitive method for detecting early gait abnormalities and assessing disability. Kadanka et al. used a standardized 10-meter walk and run test to analyze 45 DCM patients, 126 NMDCC patients, and 100 healthy controls, exploring the relationship between gait parameters and DCM-related disability. NMDCC refers to patients with spinal cord compression but no symptoms. The study found that gait analysis effectively revealed gait abnormalities in DCM patients, with 66.7% of NMDCC patients also showing abnormal gait, suggesting that gait analysis can detect early-stage cervical myelopathy beyond traditional imaging. Gait abnormalities were closely related to mJOA scores, especially in walking speed and step frequency, with more significant abnormalities in severe DCM patients. These findings highlight the importance of quantitative gait analysis in the early detection and evaluation of DCM.

By quantifying gait features, researchers not only identified significant gait abnormalities in DCM patients but also confirmed gait analysis as a valuable clinical tool. Therefore, studying gait differences between DCM patients and healthy individuals holds clinical significance, providing objective evidence for early intervention and rehabilitation planning.

3.2. Preoperative and postoperative gait differences in DCM patients: Three studies were included

Gait analysis offers unique advantages in evaluating surgical outcomes, especially in terms of objectivity and the volume of data collected, surpassing conventional subjective assessment methods such as the JOA and Nurick scales. These traditional scales often evaluate limited aspects such as sensation, motor function, and bladder function, whereas gait analysis collects diverse kinematic and dynamic data. As an objective evaluation tool, gait analysis reduces the influence of subjective bias from both the evaluator and the patient. In elderly DCM patients, gait analysis not only evaluates lower limb function but also provides insights into their quality of life. Additionally, it can guide postoperative rehabilitation by recommending specific muscle training and force application techniques, ultimately reducing bed rest time and fall risk, and improving treatment outcomes throughout the patient's lifetime.

DCM is a disease that can lead to significant functional impairment over time. It not only affects spinal health but also profoundly impacts walking ability and daily activities. Although surgical decompression is a common treatment for DCM, evaluating its effectiveness can be challenging and sometimes lacks objectivity. In this context, Haddas et al [17]. provided a new perspective by analyzing gait changes before and after surgery to understand the impact of surgery on functional recovery. In a study involving 40 DCM patients, Haddas et al. found that preoperative patients had higher ground reaction force (GRF) peaks during the push-off phase of gait, and these GRF peaks occurred earlier. Postoperatively, all GRF parameters showed improvement, although they did not reach the level of healthy controls. Specifically, postoperative patients exhibited better GRF peak values and timing, with improvements in the maximum load acceptance and intermediate support phases. However, compared to healthy individuals, postoperative patients still showed delays in the braking and propulsion phases. These improvements, combined with previous reports on gait muscle activation and gait patterns, support the notion that DCM patients experience poor postural control [18]. This study highlighted the value of detailed GRF analysis in assessing surgical outcomes for DCM patients, particularly in evaluating GRF peaks and waveforms, where surgical intervention led to significant improvements but did not fully restore normal gait patterns.

Subsequently, the research team further explored how clinical gait analysis could comprehensively assess the effects of cervical decompression surgery on gait and lower limb function in DCM patients. In a follow-up study, Haddas et al [19]. performed preoperative and postoperative comprehensive gait and lower extremity kinematic analyses on 25 DCM patients. The study found that preoperative DCM patients exhibited significant gait abnormalities, while postoperative gait parameters showed marked improvements, such as increased walking speed, step length, and step frequency, and decreased step width. Changes in lower extremity joints included increased ankle joint range of motion and decreased hip joint range of motion. Postoperatively, the hip and knee joints showed improved motion, while the pelvis and lumbar spine exhibited reduced movement. This study not only confirmed the crucial role of gait analysis in evaluating DCM surgical outcomes but also expanded our understanding of how surgical intervention impacts gait and lower limb kinematic characteristics, providing valuable insights for optimizing treatment plans and improving postoperative rehabilitation.

In a 2024 study, Mandelli et al [20]. performed 3D gait analysis on 30 DCM patients to assess the effect of surgical decompression. The results showed that severe DCM patients exhibited knee hyperflexion during the support phase, and ankle dorsiflexion was absent during the swing phase. The obtained mJOA scores correlated significantly with the knee flexion angle during the support phase. Among the patients, 15 showed improvements in kinematic gait analysis, with normal knee flexion during the support phase. This study demonstrated that the knee flexion angle during the support phase, identified through 3D gait analysis, could serve as a potential postoperative neurological recovery marker.

Since gait analysis provides kinematic and dynamic data linked to muscle activity and recovery, it, combined with spinal evoked potentials, may offer an alternative to imaging for assessing whether spinal cord compression has been fully alleviated and whether residual compression exists, guiding secondary surgical options or confirming clinical diagnoses. However, there is a lack of in-depth studies on this subject.

3.3. Gait analysis in prognostic prediction: Three studies were included

Several studies on prognostic prediction in DCM have found that preoperative JOA scores are highly correlated with postoperative improvement. Specifically, patients with better preoperative neurological function tend to have better postoperative recovery. With the objective evaluation and quantification provided by gait analysis, this preoperative neurological function can be further subdivided, which can assist in more accurately predicting patient outcomes and providing a more robust judgment for clinical treatment expectations.

Özkan et al [21]. conducted a retrospective analysis on 411 DCM patients to explore the predictive value of initial neurological symptoms for postoperative outcomes. The study found that patients with motor function deficits and gait disturbances had significantly lower preoperative mJOA scores, indicating worse baseline neurological function. Although these patients showed improvement postoperatively, their mJOA scores remained lower than those of other symptom groups. In contrast, patients with neck-arm pain were younger, had higher preoperative and postoperative scores, and showed better prognosis. This study emphasized that initial symptoms are important predictors of surgical outcomes in DCM patients and supports early surgical intervention for those with motor deficits and gait disturbances.

Trauma is an important factor in the deterioration of spinal cord function in cervical spondylosis patients, and many patients experience falls leading to fractures, further worsening spinal cord dysfunction. This phenomenon, often attributed to accidents, can also be understood as a poor prognosis caused by gait instability. Nishimura et al [22]. analyzed gait in 132 DCM patients, assessing gait features under different Nurick grades and determining gait parameters that predict fall risk. They found that even mild patients (Nurick Grade 1) showed significant abnormalities in walking speed, step length, and toe separation angle. As the Nurick grade increased, the proportion of the gait cycle spent in the support phase rose above 70%, indicating an increased fall risk. The study concluded that gait analysis can objectively assess gait stability, offering an important basis for early detection of fall risks in DCM patients and surgical intervention.

Advances in gait analysis technology have shifted from focusing purely on the collection of kinematic and dynamic data to considering gait as a whole in a three-dimensional space. Kitade et al [23]. performed 3D gait analysis on 42 DCM patients to evaluate gait characteristics of varying severity. The study found that severe patients exhibited knee hyperextension during the support phase, lacked effective tendon contraction during the swing phase, and showed significant reductions in walking speed, step length, and frequency, with shortened single support time and increased step width. Mild and moderate patients' gait characteristics were similar to those of healthy controls, but severe patients exhibited a loss of the characteristic double-peak waveform. The study also introduced analyses of muscle–tendon length and speed changes, with these indicators showing statistical correlations with mJOA scores ($P < 0.05$). Specifically, knee flexion angle was closely related to gait abnormalities, reinforcing the importance of gait analysis in evaluating DCM severity and prognosis. These findings support the application of detailed gait analysis in early intervention and treatment decision-making for DCM.

4. Discussion

Gait analysis plays a crucial role in evaluating spinal cord function and related lesions. Its non-invasive nature and ability to directly assess patients' daily activities make it a powerful tool for monitoring motor function recovery and formulating personalized treatment plans. In the future, gait analysis is expected to improve in accuracy and applicability through the use of advanced sensor technologies, data analysis methods, and artificial intelligence algorithms, thereby enhancing its role in the diagnosis, management, and treatment of spinal cord lesions [24,25].

Clinical studies on degenerative cervical myelopathy (DCM) have especially highlighted the importance of gait analysis. MRI, particularly T2-weighted imaging, reveals high signal intensity in the spinal cord of DCM patients, which is closely related to gait abnormalities [22]. EMG testing also shows muscle dysfunction, further linking it to gait [15]. While various functional assessment scales, such as the mJOA and Nurick scales, are used to evaluate DCM patients' symptoms and improvements, these conventional evaluations are more subjective and dependent on patient perceptions and clinical experience [12,23].

In contrast, gait analysis provides a more objective and quantitative evaluation. Through specific testing methods and 3D analysis techniques, clinicians can assess lower-limb range of motion, muscle strength, and balance in DCM patients. This method captures a range of spatiotemporal, kinematic, and dynamic parameters, which have been proven to be impaired in DCM patients [12,23].

Despite the deep insights gait analysis offers, its application is limited by the high cost of equipment, operational complexity, and the need for specialized training. Therefore, developing standardized, user-friendly, and cost-effective gait analysis tools is crucial. Recently, smartphone-based gait analysis applications have shown potential in managing chronic diseases by analyzing motion sensor data and providing real-time information on patient acceleration, joint mobility, and more [26,27]. Given that DCM predominantly affects elderly patients who often have limited medical access and delayed intervention, integrating gait analysis with everyday wearables such as smartphones and watches for cervical disease screening and early clinical intervention would have significant potential for wide-scale application and substantial value throughout a patient's life cycle.

The team at the Chinese University of Hong Kong has developed a system based on the 3D-Move AI algorithm, which uses smartphone cameras to capture 2D images and reconstruct human three-dimensional movement. They have also developed a smart rehabilitation guidance system via an app to guide exercise. Similar technology could be used for early home screening of gait abnormalities. However, the standardization of this technology remains a current challenge, and the application of gait analysis in cervical disease diagnosis must be standardized gradually. This will require parallel progress in the "device-protocol-parameter-interpretation" framework, with the ultimate goal of creating a mature system, similar to current MRI reading practices, complete with standardized scanning sequences, measurement methods, and diagnostic terminology [28,29].

In conclusion, the importance of gait analysis as a tool for evaluating spinal lesions continues to grow. It not only shows potential in the diagnosis and functional evaluation of DCM patients but also aids in monitoring treatment effects and long-term management. Previous studies support the use of gait analysis as a sensitive method for identifying early gait abnormalities in cervical myelopathy patients and assessing their prognosis. It significantly reveals the dynamic improvements in gait and lower limbs after cervical decompression surgery, providing crucial information for understanding the effects of surgical interventions and optimizing patients' rehabilitation treatment. In the future, with the development of technology and the introduction of new methods, gait analysis is expected to better serve patients' rehabilitation processes and become an indispensable part of personalized medical services.

5. Conclusion

In summary, this study shows that quantitative gait analysis can sensitively identify early gait abnormalities in patients with degenerative cervical myelopathy (DCM). These abnormalities are characterized by decreased spatiotemporal parameters, such as walking speed, step frequency, and step length, along with reduced range of motion in the knee and ankle joints and abnormal ground reaction force patterns. After cervical decompression surgery, these parameters show significant improvement, although they still differ from healthy individuals, indicating that gait analysis can serve as an important objective indicator for evaluating surgical outcomes. Additionally, gait parameters correlate significantly with functional scores and are valuable in predicting surgical prognosis and assessing fall risks. With the development of wearable sensors and artificial intelligence, quantitative gait analysis is expected to become an indispensable tool in DCM diagnosis and treatment systems, providing objective and quantifiable data for early screening, surgical decision-making, rehabilitation guidance, and long-term follow-up, thus improving the overall clinical management quality of DCM patients.

CRediT authorship contribution statement

Jiuheng Li: Conceptualization, Data curation, Investigation, Methodology, Software, Writing – original draft. **Yuetao Zhu:** Data curation, Investigation, Writing – review & editing. **Ning Zhang:** Data curation, Formal analysis, Methodology. **Qinpeng Song:** Formal analysis, Investigation, Methodology. **Jinchao Wang:** Investigation, Project administration, Validation. **Chao Wang:** Conceptualization, Methodology, Supervision, Writing – original draft. **Xiao Han:** Conceptualization, Investigation, Methodology, Supervision, Writing – original draft.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Xiao Han reports financial support was provided by Beijing Natural Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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